Wenxue Li

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RESEARCH INTERESTS

I am broadly interested in datacenter networking, with a particular focus on high-speed RDMA networking, distributed AI systems, and networked systems for AI.

WORK EXPERIENCE

Senior Engineer (Alibaba Star), Alibaba Group *2026 – Present*
Shenzhen, China

EDUCATION

Hong Kong University of Science and Technology, Hong Kong SAR *2021 – 2026*
Ph.D. in Computer Science and Engineering
Thesis: Towards Efficient and Scalable RDMA Networking for Datacenter Applications
Advisor: Prof. Kai Chen

Zhejiang University, Hangzhou, China *2016 – 2021*
B.Sc. in Electronic Science and Technology

PUBLICATIONS

* indicates co-first authors; # indicates corresponding authors

Conference Publications:

[C19] **Wenxue Li**, Zihao Wang, Zhaoxiang Bao, Zhenghang Ren, Yijun Sun, Bowen Liu, Han Tian, Junxue Zhang, Xiaolin Chen, Kai Chen. Pyxis: Unlocking Multi-NIC Bandwidth for Incast Traffic in AI Clusters. In the 24th USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Providence, RI, USA, 11–13 May, 2027.

[C18] Yu Zhu, Xinhao Yang, Zhaoshi Li, Baodong Wu, Boxun Li, **Wenxue Li**, Guohao Dai, Hongyang Jia, Shuwen Deng, Huazhong Yang, Yuan Xie, Zhenhua Zhu, Yu Wang. Making Protocol Effects Explicit: LLM-to-Packet Simulation for Superpod Interconnect Exploration. In the 33rd IEEE International Symposium on High-Performance Computer Architecture (**HPCA**), Salt Lake City, UT, USA, 20–24 March, 2027.

[C17] Junxue Zhang, Han Tian, Xinyang Huang, **Wenxue Li**, Kaiqiang Xu, Dian Shen, Kai Chen. FCP: Fast User-space RDMA Control Plane Setup for Elastic Computing. In the 17th ACM Symposium on Cloud Computing (**SoCC**), Singapore, 18–20 November, 2026.

[C16] Han Tian, Han Wang, Wenbo Li, Xudong Liao, Decang Sun, **Wenxue Li**, Donghui Chen, Bin Huan, Senbo Fu, Junxue Zhang, Dian Shen, Kai Chen. PolicyCache: Intra-flow Learning in Congestion Control. In the 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Renton, WA, USA, 4–6 May, 2026.

[C15] Xiangzhou Liu, **Wenxue Li#**, Zihao Wang, Kai Chen#. Enabling Packet Spraying over Commodity RNICs with In-Network Support. In the 21st ACM European Conference on Computer Systems (**EuroSys**), Edinburgh, UK, 27–30 April, 2026.

[C14] Yunxuan Zhang, Hao Wang, Han Tian, Liu Yang, Xudong Liao, **Wenxue Li**, Ping Yin, Bowen Liu, Kai Chen. MFS: An Efficient Model Family Serving System for Large Language Models. In the 21st ACM European Conference on Computer Systems (**EuroSys**), Edinburgh, UK, 27–30 April, 2026.

[C13] Han Tian, Wenbo Li, Junxue Zhang, Xudong Liao, Decang Sun, Donghui Chen, Bin Huang, **Wenxue Li**, Yong Wang, Kai Chen. Learn-to-Probe: Achieving Signal Distinguishability in Learning-based Congestion Control. In the 21st ACM European Conference on Computer Systems (**EuroSys**), Edinburgh, UK, 27–30 April, 2026.

[C12] Xudong Liao, Yijun Sun, Han Tian, Xinchen Wan, Yilun Jin, Zilong Wang, Zhenghang Ren, Xinyang Huang, **Wenxue Li**, Kin Fai Tse, Zhizhen Zhong, Guyue Liu, Ying Zhang, Xiaofeng Ye, Yiming Zhang, Kai Chen. MixNet: A Runtime Reconfigurable Optical-Electrical Fabric for Distributed Mixture-of-Experts Training. In the 39th annual conference of the ACM Special Interest Group on Data Communication (**SIGCOMM**), Portugal, September 8–11, 2025.

[C11] **Wenxue Li**, Xiangzhou Liu, Yunxuan Zhang, Zihao Wang, Wei Gu, Tao Qian, Gaoxiong Zeng, Shoushou Ren, Xinyang Huang, Zhenghang Ren, Bowen Liu, Junxue Zhang, Bingyang Liu, Kai Chen. Revisiting RDMA Reliability for Lossy Fabrics. In the 39th annual conference of the ACM Special Interest Group on Data Communication (**SIGCOMM**), Portugal, September 8–11, 2025. **[Best Student Paper Award Honorable Mention]**

[C10] Bowen Liu, Xinyang Huang, Qijing Li, Zhuobin Huang, Yijun Sun, **Wenxue Li**, Junxue Zhang, Ping Yin, Kai Chen. "CEIO: A Cache-Efficient Network I/O Architecture for NIC-CPU Data Paths." In the 39th annual conference of the ACM Special Interest Group on Data Communication (**SIGCOMM**), Portugal, September 8–11, 2025.

[C9] Xiangzhou Liu, **Wenxue Li**, Kai Chen. Enabling Packet Spraying over Commodity RNICs with In-Network Support. In the 9th Asia-Pacific Workshop on Networking (**APNet**), Shanghai, China, August 7–8, 2025.

[C8] Yuxuan Li, Zhenghang Ren, **Wenxue Li**, Xiangzhou Liu, Kai Chen. Congestion Control for AI Workloads with Message-Level Signaling. In the 9th Asia-Pacific Workshop on Networking (**APNet**), Shanghai, China, August 7–8, 2025.

[C7] Jinbin Hu*, **Wenxue Li***, Xiangzhou Liu, Junfeng Wang, Bowen Liu, Ping Yin, Jianxin Wang, Jiawei Huang, Kai Chen. "FLB: Fine-grained Load Balancing for Lossless Datacenter Networks." In the 2025 USENIX Annual Technical Conference (**ATC**), Boston, MA, USA, July 2025.

[C6] Zhenghang Ren, Yuxuan Li, Zilong Wang, Xinyang Huang, **Wenxue Li**, Kaiqiang Xu, Xudong Liao, Yijun Sun, Bowen Liu, Han Tian, Junxue Zhang, Mingfei Wang, Zhizhen Zhong, Guyue Liu, Ying Zhang, Kai Chen. "Enabling Efficient GPU Communication over Multiple NICs with FuseLink." In the 19th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), Boston, MA, USA, July 2025.

[C5] **Wenxue Li**, Xiangzhou Liu, Yuxuan Li, Yilun Jin, Han Tian, Zhizhen Zhong, Guyue Liu, Ying Zhang, Kai Chen. Understanding Communication Characteristics of Distributed Training. In the 8th Asia-Pacific Workshop on Networking (**APNet**), Sydney, Australia, August 2024.

[C4] **Wenxue Li***, Junyi Zhang*, Yufei Liu, Gaoxiong Zeng, Zilong Wang, Chaoliang Zeng, Pengpeng Zhou, Qiaoling Wang, Kai Chen. "Cepheus: Accelerating Datacenter Applications with High-Performance RoCE-Capable Multicast." In the 30th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**), Edinburgh, Scotland, March 2024.

[C3] **Wenxue Li**, Chaoliang Zeng, Jinbin Hu, Kai Chen. "Towards Fine-grained and Practical Flow Control for Datacenter Networks." In the 31st IEEE International Conference on Network Protocols (**ICNP**), Reykjavik, Iceland, October 2023.

[C2] Zilong Wang, Layong Luo, Qingsong Ning, Chaoliang Zeng, **Wenxue Li**, Xinchen Wan, Peng Xie, Tao Feng, Ke Cheng, Xiongfei Geng, Tianhao Wang, Weicheng Ling, Kejia Huo, Pingbo An, Kui Ji, Shideng Zhang, Bin Xu, Ruiqing Feng, Tao Ding, Kai Chen, Chuanxiong Guo. "SRNIC: A Scalable Architecture for RDMA NICs." In the 20th USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Boston, MA, USA, April 2023.

[C1] **Wenxue Li**, Chaoliang Zeng, Jinbin Hu, Kai Chen. Scaling Switch-driven Flow Control with Aquarius. In the 7th Asia-Pacific Workshop on Networking (**APNet**), Hong Kong SAR, June 2023.

Journal Publications:

[J5] **Wenxue Li**, Xiangzhou Liu, Yuxuan Li, Yilun Jin, Han Tian, Zhenghang Ren, Xudong Liao, Zhizhen Zhong, Guyue Liu, Ying Zhang, Kai Chen. Understanding Communication Predictability of Distributed LLM Training. In IEEE Transactions on Networking (**ToN**), 2026.

[J4] **Wenxue Li**, Xiangzhou Liu, Yunxuan Zhang, Zihao Wang, Wei Gu, Tao Qian, Gaoxiong Zeng, Shoushou Ren, Xinyang Huang, Zhenghang Ren, Bowen Liu, Junxue Zhang, Bingyang Liu, Kai Chen. Reliable RDMA over Lossy Fabrics via Data-Control Partitioning. In IEEE/ACM Transactions on Networking (**ToN**), 2026.

[J3] Jinbin Hu, Siyao Li, **Wenxue Li**, Xiangzhou Liu, Bowen Liu, Ping Yin, Mengyu Ma, Jin Wang, Jianxin Wang, Jiawei Huang, Kai Chen. Towards Fine-grained Load Balancing with Congested-flow Isolation in Lossless Datacenters. In IEEE/ACM Transactions on Networking (**ToN**), 2026.

[J2] **Wenxue Li**, Junyi Zhang, Yufei Liu, Gaoxiong Zeng, Zilong Wang, Chaoliang Zeng, Pengpeng Zhou, Qiaoling Wang, Kai Chen. High-Performance RoCE-Capable Multicast for Commodity RDMA Datacenters. In IEEE/ACM Transactions on Networking (**ToN**), 2025.

[J1] **Wenxue Li**, Chaoliang Zeng, Jinbin Hu, Kai Chen. FlowSail: Fine-grained and Practical Flow Control for Datacenter Networks. In IEEE/ACM Transactions on Networking (**ToN**), 2024.

HONORS & AWARDS

- CSE Best PhD Dissertation Award (2025–26), HKUST 2026
- Rising Stars Women in Engineering (Shortlisted) 2026
- SIGCOMM Best Student Paper Award Honorable Mention 2025
- SIGCOMM 2025 N2Women Travel Grants 2025
- ATC 2025 Travel Grants 2025
- HPCA 2024 Travel Grants 2024
- Postgraduate Studentship, HKUST 2021–2025
- First-class Scholarship, ZJU 2019
- Zhejiang Daily - Alibaba New Media Scholarship, ZJU 2019
- Outstanding Student Leader, ZJU 2018

INVITED TALKS

- Revisiting RDMA Reliability for Lossy Fabrics
  ▷ Seminar @ Best Paper Deep Dive, Huang Danian Chaspark 01/2026
  ▷ Seminar @ ByteDance 10/2025
- Towards Efficient and Scalable RDMA Networking for Datacenter Applications 12/2025
  ▷ Seminar @ University of Science and Technology of China

PROFESSIONAL SERVICES

Conference Committees

INFOCOM Technical Program Committee 2027
SIGCOMM Artifact Evaluation Committee 2026

Teaching Assistant

COMP3511 Operating Systems 2025 Spring
COMP5621 Computer Networks 2023 Fall
COMP3511 Operating Systems 2022 Fall
COMP4651 Cloud Computing and Big Data Systems 2022 Spring

OTHER EXPERIENCES

Research Intern, Huawei Hong Kong Research Center June 2024 – July 2025
Hong Kong SAR

Software Engineer Intern, ByteDance July 2020 – June 2021
Hangzhou, China